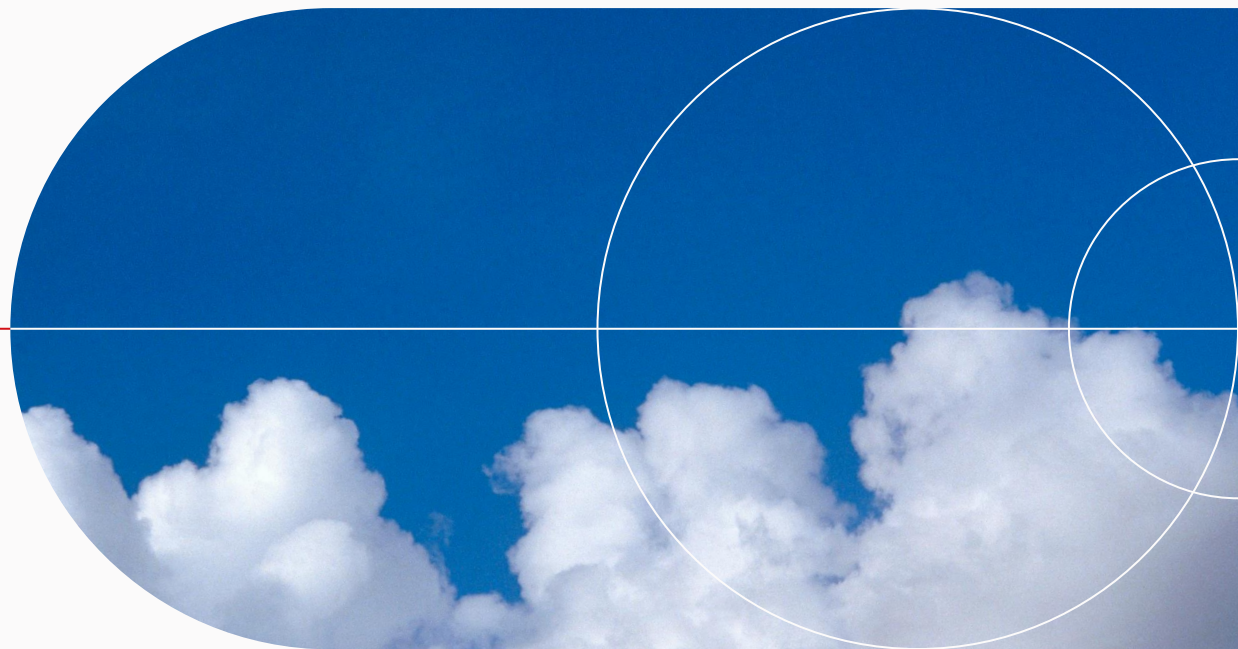


Detecting Breathing Patterns *Project Proposal*

Machine Learning and Sensing

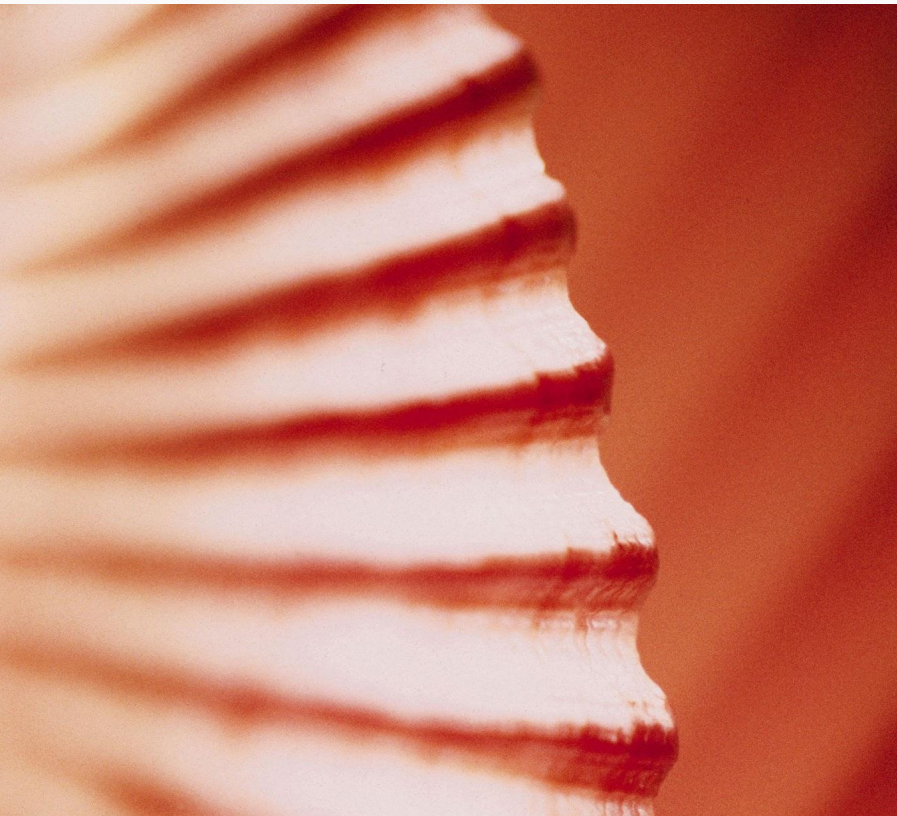
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Agenda

SPRING 2026

- 01 Motivation
- 02 Sensing Component
- 03 Machine Learning Component
- 04 Evaluation
- 05 Live Demo Setup



Motivation

Background

- Breathing patterns reflect stress, fatigue, or respiratory discomfort
- Abnormal respiratory rate - early and reliable sign of cardiac arrest

Current Solutions

- Clinical monitoring is expensive, lab-based, and inaccessible
- Most wearables track heart rate but ignore respiratory patterns

Our Approach

- Real-time detection enabling early intervention before symptoms escalate

Sensing Component

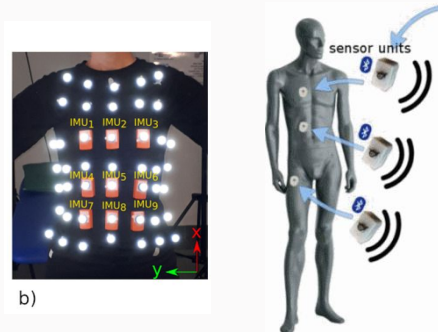
Using both an IMU and PPG to due to correlation of cardiovascular with respiratory system.

Inertial Measurement Unit (IMU)

Monitor respiratory motion

- using combo of accelerometers and gyroscopes (detecting changes in pitch, yaw, roll)

Placement → multiple IMUs at various points (most likely the chest)



Photoplethysmography (PPG)

Estimates heart rate

- using a light source and photodetector at the skin surface
- noninvasive, optical measurements

Placement → wrists (like in smartwatches)

Challenge: Detecting breathing patterns during high activity.

Machine Learning Component

Goal: Classify breathing states → Multimodal Time-Series Classification

Model: Ensemble Modeling (Random Forest*) or Logistic Regression

*Generally good for noisy sensor data

Input: IMU → Respiratory Motion, PPG → Cardiovascular activity

Output: Classify into 1 of 2 categories: Normal/Resting vs Irregular

Examples of Extracted Features:

- Mean acceleration
- Pulse amplitude
- Heart rate (BPM)
- Interbeat Interval
- Heart rate variability

Evaluation

Goal: Reliable detection of irregular breathing in real-world conditions

Model Validation

- Train/Test Split (80/20)
- Ensures generalization to unseen data

Performance Metrics

- Accuracy
- Precision & Recall
- F1 Score

Error Analysis

- Confusion Matrix
- Identify missed irregular breathing cases

Baseline Comparison

- Compare vs simple classifier (e.g., always “normal”)
- Demonstrates added model value

Real-Time Performance

- Latency of predictions
- Stability over continuous sensing

Live Demo Setup

Gist: Recording of a live actor using the sensor with frames of real-time serial plot in Arduino detecting abnormal patterns.

“Behavior–Signal Correlation” - This shows that the system is used to interpret how actions influence sensor output.



Source: Arduino Docs

